

① Construct three address code for the following
 $(a < b) \text{ \& } (c < d) \text{ or } (a > d)$ then

$$t = x + y * t$$

$$\text{else } z = z + 1$$

37/40

Step 1: condition is: $a < b$
 $c < d$
 $a > d$

the structured is $(a < b) \text{ and } (c < d) \text{ or } (a > d)$

Step 2: create table. $\rightarrow L_1, L_2, L_3, L_4, L_5$

Step 3: Generate TAC for the condition

L_1 : if $a < b$ goto $L_2 \rightarrow$ goto L_5

L_2 : if $a < c < d$ goto $L_3 \rightarrow$ goto L_5

L_3 : if $a > d$ goto $L_4 \rightarrow$ goto L_5 .

Step 4: Generate TAC for the then part

$$z = x + y * z.$$

$$L_{11}: t_1 = y * z.$$

$$t_2 = x + t_1$$

$$t = t_2 \rightarrow \text{goto } L_6$$

Else part

$$i) \quad z = z + 1$$

$$L_7: t_1 = z + 1$$

$$z = t_1$$

Final: L_1 : if $a < b$ goto L_2
 goto L_5

L_2 : if $c < d$ goto L_3
 goto L_5

L_3 : if $a > d$ goto $L_4 \rightarrow$ goto L_5 .

$L_1: L_1 = Y * E.$

$L_2: X + E_1$

$Z = E_2 \rightarrow \text{goto } L_6$

② translate $(a < b) \& (c < d)$ or $(e < f)$ into three address statements using back patching

① Step 1: condition is: $a < b, c < d, e < f$

the structure is $(a < b) \& (c < d)$ or $(e < f)$

Step 2: if $a < b$ goto -

goto -
if $c < d$ goto -
goto -

if $e < f$ goto -
goto -

Step 3: handle AND

for $(a < b) \& (c < d)$ if $a < b$ is true:

if $a < b$ goto 3

Step 4: final back patching code

if $a < b$ goto 3
goto 8

if $c < d$ goto 2

goto 8
if $e < f$ goto 9
goto 1

Tree, Patch

③ Construct three address code using the following statement for evaluation for between $(a < b)$ and

$(c < d)$ or $(a < f)$.

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B \& B_2$

$B \rightarrow ! B,$

$B \rightarrow G, \text{ not } G,$

$B_1 \rightarrow \text{true}, B \rightarrow \text{false}$

1) Step 1: $a > b, c < d, e < f$

Assume Expression is:

$(a > b) \wedge (c < d) \wedge (e < f)$

Step 2: Generate three address code for $a > b$

if $a > b$ goto true
goto L_1

Step 3: $c < d$

if $c < d$ goto L_1 true
goto L_2

Step 4: $e < f$

if $e < f$ goto true.
goto (False)

true, false

true: $\wedge$ Boolean expression is true
~~goto~~ LEAD

false: $\wedge$ Boolean expression is false.

5) Generate three address code for

the following program fragment.

while $(A \neq 0 \wedge B > 0)$ do

if $A = 1$ then $C = C + 1$

repeat while $A \neq 0$ do

$A = A + B$

A) C_1 : if $A < C$ goto L_1
goto C_{end}
 L_1 : if $B > D$ goto L_2
goto C_{end}

L_2 : if $A \geq 1$ goto L_{u1}
goto L_r

C_{u1} : $C_1 = C + 1$
 $C = L_1$
goto L_1

L_r : if $A \leq D$ goto L_c
goto L_1

L_c : $L_2 = A + B$
 $A = L_2$
goto //